

A Technical Introduction to Protein structure prediction

Section 1

In the case of complexes of two or more proteins, where the structures of the proteins are known or can be predicted with high accuracy, protein–protein docking methods can be used to predict the structure of the complex. Information of the effect of mutations at specific sites on the affinity of the complex helps to understand the complex structure and to guide docking methods.

Section 2

The practical role of protein structure prediction is now more important than ever. Massive amounts of protein sequence data are produced by modern large-scale DNA sequencing efforts such as the Human Genome Project. Despite community-wide efforts in structural genomics, the output of experimentally determined protein structures—typically by time-consuming and relatively expensive X-ray crystallography or NMR spectroscopy—is lagging far behind the output of protein sequences. The protein structure prediction remains an extremely difficult and unresolved undertaking. The two main problems are the calculation of protein free energy and finding the global minimum of this energy. A protein structure prediction method must explore the space of possible protein structures which is astronomically large. These problems can be partially bypassed in "comparative" or homology modeling and fold recognition methods, in which the search space is pruned by the assumption that the protein in question adopts a structure that is close to the experimentally determined structure of another homologous protein. In contrast, the *de novo* protein structure prediction methods must explicitly resolve these problems. The progress and challenges in protein structure prediction have been reviewed by Zhang.

Section 3

AI methods AlphaFold was one of the first AIs to predict protein structures. It was introduced by Google's DeepMind in the 13th CASP competition, which was held in 2018. AlphaFold relies on a neural network approach, which directly predicts the 3D coordinates of all non-hydrogen atoms for a given protein using the amino acid sequence and aligned homologous sequences. The AlphaFold network consists of a trunk which processes the inputs through repeated layers, and a structure module which introduces an explicit 3D structure. Earlier neural networks for protein structure prediction used LSTM.

Section 4

Modeling of side-chain conformations Accurate packing of the amino acid side chains represents a separate problem

in protein structure prediction. Methods that specifically address the problem of predicting side-chain geometry include dead-end elimination and the self-consistent mean field methods. The side chain conformations with low energy are usually determined on the rigid polypeptide backbone and using a set of discrete side chain conformations known as "rotamers". The methods attempt to identify the set of rotamers that minimize the model's overall energy. These methods use rotamer libraries, which are collections of favorable conformations for each residue type in proteins. Rotamer libraries may contain information about the conformation, its frequency, and the standard deviations about mean dihedral angles, which can be used in sampling. Rotamer libraries are derived from structural bioinformatics or other statistical analysis of side-chain conformations in known experimental structures of proteins, such as by clustering the observed conformations for tetrahedral carbons near the staggered (60° , 180° , -60°) values. Rotamer libraries can be backbone-independent, secondary-structure-dependent, or backbone-dependent. Backbone-independent rotamer libraries make no reference to backbone conformation, and are calculated from all available side chains of a certain type (for instance, the first example of a rotamer library, done by Ponder and Richards at Yale in 1987). Secondary-structure-dependent libraries present different dihedral angles and/or rotamer frequencies for α -helix, β -sheet, or coil secondary structures. Backbone-dependent rotamer libraries present conformations and/or frequencies dependent on the local backbone conformation as defined by the backbone dihedral angles ϕ and ψ , regardless of secondary structure. The modern versions of these libraries as used in most software are presented as multidimensional distributions of probability or frequency, where the peaks correspond to the dihedral-angle conformations considered as individual rotamers in the lists. Some versions are based on very carefully curated data and are used primarily for structure validation, while others emphasize relative frequencies in much larger data sets and are the form used primarily for structure prediction, such as the Dunbrack rotamer libraries. Side-chain packing methods are most useful for analyzing the protein's hydrophobic core, where side chains are more closely packed; they have more difficulty addressing the looser constraints and higher flexibility of surface residues, which often occupy multiple rotamer conformations rather than just one.

Section 5

Comparative protein modeling Comparative protein modeling uses previously solved structures as starting points, or templates. This is effective because it appears that although the number of actual proteins is vast, there is a limited set of tertiary structural motifs to which most proteins belong. It has been suggested that there are only

around 2,000 distinct protein folds in nature, though there are many millions of different proteins. The comparative protein modeling can combine with the evolutionary covariation in the structure prediction. These methods may also be split into two groups:

variation in multiple sequence alignments is taken into account.

Section 6

Protein structure prediction is the inference of the three-dimensional structure of a protein from its amino acid sequence—that is, the prediction of its secondary and tertiary structure from primary structure. Structure prediction is different from the inverse problem of protein design. Protein structure prediction is one of the most important goals pursued by computational biology and addresses Levinthal's paradox. Accurate structure prediction has important applications in medicine (for example, in drug design) and biotechnology (for example, in novel enzyme design). Starting in 1994, the performance of current methods is assessed biennially in the Critical Assessment of Structure Prediction (CASP) experiment. A continuous evaluation of protein structure prediction web servers is performed by the community project Continuous Automated Model EvaluatiOn (CAMEO3D).

Section 7

Since AlphaFold outputs protein coordinates directly, AlphaFold produces predictions in graphics processing unit (GPU) minutes to GPU hours, depending on the length of protein sequence. The European Bioinformatics Institute together with DeepMind have constructed the AlphaFold – EBI database for predicted protein structures.

Section 8

β -sheets are formed by H-bonds between an average of 5–10 consecutive amino acids in one portion of the chain with another 5–10 farther down the chain. The interacting regions may be adjacent, with a short loop in between, or far apart, with other structures in between. Every chain may run in the same direction to form a parallel sheet, or in the reverse direction to form an antiparallel sheet, or the chains may form a mixed sheet. The pattern of hydrogen bonding is different in the parallel and antiparallel configurations. Each amino acid in the interior strands of the sheet forms two H-bonds with neighboring amino acids, whereas each amino acid on the outside strands forms only one bond with an interior strand. Looking across the sheet at right angles to the strands, more distant strands are rotated slightly counterclockwise to form a left-handed twist. The $C\alpha$ -atoms alternate above and below the sheet in a pleated structure, and the R side groups of the amino acids alternate above and below the pleats. The Φ and Ψ angles of the amino acids in sheets vary considerably in one region of the Ramachandran plot. It is more difficult to predict the location of β -sheets than of α -helices. The situation improves somewhat when the amino acid